

OFFICIAL AGROSYNC TECHNICAL SPECIFICATION

Software Requirements Specification (SRS)

AgroSync Farm Management System

Version: 4.0

SDLC Stage: Production Baseline & Comprehensive Feature Specification

Status: Approved

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete, authoritative specification of the functional, non-functional, security, compliance, and architectural requirements for the **AgroSync** (formerly SmartFarm) Agricultural Enterprise Management Platform. It serves as the primary contract and blueprint for software engineering, quality assurance, system administration, and operational auditing.

1.2 Scope of the System

AgroSync is an end-to-end, multi-tenant agricultural enterprise resource planning (ERP) system designed for commercial farming enterprises, cooperative unions, and agricultural estates. The platform unifies:

Executive Governance & Multi-Category Financial Control: Global budgeting, sector delegation, and dual-ledger Accounts Payable (AP) and Accounts Receivable (AR) management for **Farm Administrators**.

Sector Planning & Procurement Management: Crop and livestock seasonal cycles, input supplier procurement, inventory replenishment, and field supervisor portfolio management for **Farm Managers**.

Field Operations, Agronomic Scheduling & Harvest Control: Field task scheduling with automated agronomic presets, harvest produce logging, farm supply consumption, and verified legal labor task allocation for **Field Supervisors**.

Produce Sales & Customer Credit Governance: Farm-gate sales validation ensuring sales cannot exceed harvested stocks, credit limit enforcement, partial down-payments, and chronological installment payment audit trails.

Supplier Accounts Payable & Warehouse Restocking: Purchase order logging, optional direct warehouse restocking, and voucher settlement audit trails.

Proactive Notification Engine: Real-time cross-cutting alerts for pending user approvals, critical low inventory thresholds, upcoming/overdue field activities, and outstanding trade balances.

Executive Analytics & Unified Ledger: Real-time KPI cards (Cash Received in Bank vs Uncollected Customer Debts vs Gross Booked Revenue), interactive Sector Revenue & 2-Status Project Donut Charts, and server-side paginated transaction feeds.

1.3 Target Audience & User Roles

- Farm Administrator (ADMIN):** Supreme authority with global governance, user lifecycle approval, quota enforcement, sector creation, budget oversight, and complete financial visibility.
- Farm Manager (MANAGER):** Assigned to specific farm sectors/categories. Manages project creation, budget allocations, input procurement, supplier relations, and supervisor task assignment.
- Field Supervisor (SUPERVISOR):** Assigned to supervise specific field projects. Coordinates daily field tasks, schedules agronomic operations, logs harvests, records inventory usage, and allocates verified employee labor with a strict financial shield (cost amounts and company bank balances are hidden).

- 4. **Verified Farm Workers (EMPLOYEES)**: Registered labor force members with national ID and age verification (\$Age \ge 18\$) allocated to field activities with computed wage entitlements.
- 5. **Produce Customers / Offtakers (CUSTOMERS)**: Registered institutional buyers, wholesalers, and retailers purchasing farm produce on cash or credit terms.
- 6. **Input & Feed Suppliers (SUPPLIERS)**: Commercial vendors supplying seeds, fertilizers, feeds, chemicals, veterinary drugs, and equipment.

1.4 Definitions, Acronyms, and Abbreviations

AP (Accounts Payable): Cumulative financial liabilities owed by the farm to input suppliers.

AR (Accounts Receivable): Cumulative outstanding credit balances owed to the farm by produce buyers.

FIFO (First-In, First-Out): Debt settlement algorithm applying unallocated payments against the oldest outstanding invoices first.

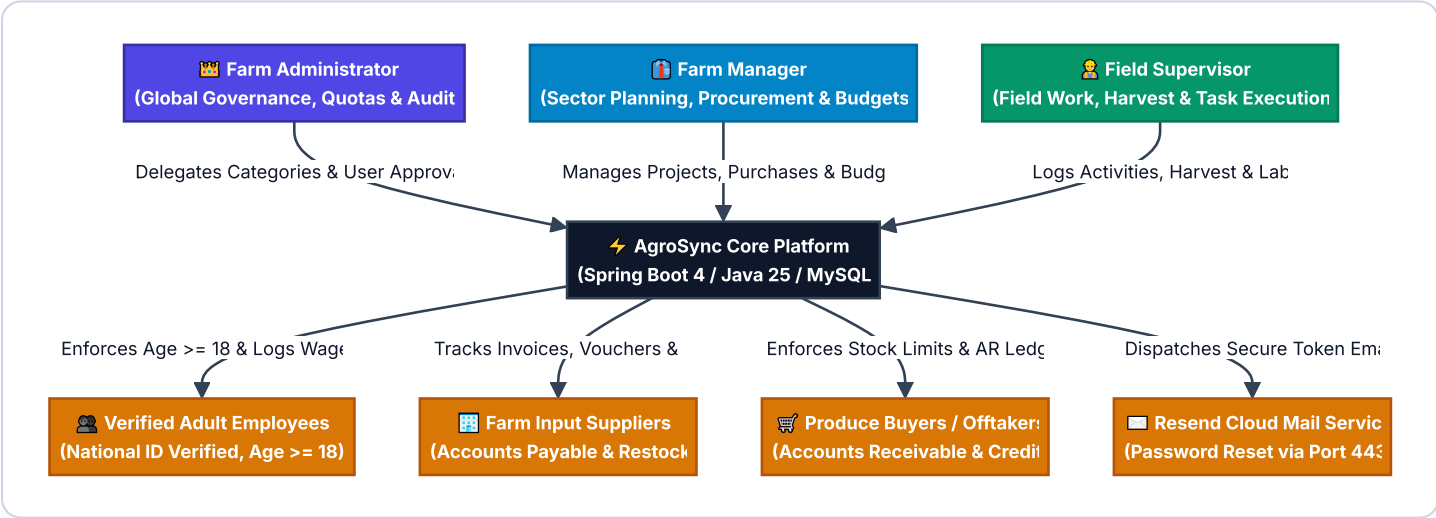
PBAC (Privilege-Based Access Control): Granular permission flags assigned to managers and supervisors (CAN_MANAGE_BUDGETS , CAN_RECORD_SALES , CAN_RECORD_EXPENSES , CAN_LOG_ACTIVITIES , CAN_RECORD_HARVEST , CAN_USE_INVENTORY , CAN_VIEW_FINANCIALS).

RBAC (Role-Based Access Control): Role hierarchy enforcing ADMIN > MANAGER > SUPERVISOR .

Resend HTTP API: HTTPS RESTful email service (port 443) used for password recovery and system transactional communications.

2. System Architecture & Context

2.1 System Context Diagram



3. Detailed Functional Requirements

3.1 Authentication, User Governance & RBAC (FR-AUTH)

- FR-AUTH-1 (Zero-State Admin Bootstrap)**:
- When the database contains zero registered users (\$Count = 0\$), the registration workflow automatically provisions the first registrant as the **Primary System Administrator** with **ACTIVE** status and immediate administrative access.
- FR-AUTH-2 (Approval Lifecycle)**:
- All subsequent user registrations default to **PENDING_APPROVAL** . Users cannot access protected farm resources until approved and assigned an operational role by an Administrator or authorized Manager.
- FR-AUTH-3 (Role Quotas & Capacity Limits)**:

- System enforces a strict maximum quota of **2 Farm Managers** system-wide.
- System enforces a strict maximum quota of **10 Field Supervisors** system-wide.
- Each Field Supervisor has a configurable maximum active project capacity (`maxProjectCapacity` , default \$4\$ projects).

FR-AUTH-4 (Granular Privilege Delegation - PBAC):

Administrators can assign discrete privileges to Managers and Supervisors:

- `CAN_MANAGE_BUDGETS` : Authorizes editing project budgets and configurations.
- `CAN_RECORD_SALES` : Authorizes farm-gate sales recording.
- `CAN_RECORD_EXPENSES` : Authorizes logging operating expenses.
- `CAN_LOG_ACTIVITIES` : Authorizes scheduling and completing field tasks.
- `CAN_RECORD_HARVEST` : Authorizes logging crop and livestock harvests.
- `CAN_USE_INVENTORY` : Authorizes deducting supplies from warehouse stock.
- `CAN_VIEW_FINANCIALS` : Authorizes viewing monetary values, budget gauges, and cost figures.

FR-AUTH-5 (Self-Service Password Recovery via Resend):

Users can initiate password resets via email. The system generates a cryptographic reset token valid for **15 minutes** and delivers the recovery link via Resend HTTP API (Port 443 HTTPS) with SMTP fallback.

3.2 Sector & Category Management (FR-SEC)

FR-SEC-1 (Sector Hierarchy):

The system organizes all farm production under Sectors/Categories (e.g., *Crops & Horticulture*, *Dairy Cattle*, *Poultry*, *Aquaculture*, *Livestock*).

FR-SEC-2 (Manager Sector Assignment):

Farm Managers are assigned to one or more sectors (`assignedCategories`). Managers only have visibility and operational authority over projects within their assigned sectors.

FR-SEC-3 (Sector Financial & Operational Aggregation):

Each sector aggregates cumulative budgets, expenses incurred, produce sales generated, harvests collected, and active projects in real-time.

3.3 Project Lifecycle Management (FR-PROJ)

FR-PROJ-1 (Streamlined Two-Status Model):

All farm projects strictly follow a two-state lifecycle:

1. `Active` : Currently ongoing field project accepting activity logs, expense records, harvest entries, supply usage, and sales.
2. `Completed` : Concluded project cycle preserved in historical archive with finalized financial net yield.

(Legacy statuses like `in_progress` , `done` , `inactive` , or `pending` are automatically normalized into `active` or `completed`).

FR-PROJ-2 (Project Scoping & Supervisor Assignment):

Each project is linked to a parent Sector/Category and assigned to a specific Field Supervisor. Supervisors can only view and manage projects in their portfolio.

FR-PROJ-3 (Supervisor Financial Shield):

For Field Supervisors without the `CAN_VIEW_FINANCIALS` privilege, all monetary metrics (budget amount, expenses total, sales revenue, net value, unit prices) are masked to `$0.00` / hidden to protect farm commercial confidentiality.

FR-PROJ-4 (Project Dashboard Tabbed Records):

The Project Dashboard provides structured tabs for:

- **Expenses:** Direct operational costs with title, unit price, quantity, amount, and notes.
- **Activities:** Scheduled agronomic operations with priority badges, status toggles, and labor assignments.
- **Harvest:** Produce volume logged with units of measurement.
- **Sales:** Farm-gate sales records linked to registered customers.
- **Supplies:** Warehouse inventory materials consumed directly by this project.

3.4 Field Operations, Agronomic Scheduling & Labor (FR-OPS)

FR-OPS-1 (Agronomic Task Scheduling Presets):

Field supervisors can schedule agronomic tasks with one-click automated date calculation presets:

- **+2 Wks (Spraying)** : Sets Scheduled Date = Today \$+ 14\$ days, Priority = **HIGH** , Type = **Field Care** .
- **+3 Wks (Weeding)** : Sets Scheduled Date = Today \$+ 21\$ days, Priority = **MEDIUM** , Type = **Field Care** .
- **+4 Wks (Top-Dressing)** : Sets Scheduled Date = Today \$+ 28\$ days, Priority = **HIGH** , Type = **Fertilization** .
- **+2 Mos (Harvesting)** : Sets Scheduled Date = Today \$+ 60\$ days, Priority = **URGENT** , Type = **Harvesting** .

FR-OPS-2 (Task Lifecycle & Countdown Badging):

Tasks support statuses (**SCHEDULED** , **IN_PROGRESS** , **COMPLETED** , **CANCELLED**) and priorities (**LOW** , **MEDIUM** , **HIGH** , **URGENT**). Cards dynamically render status badges: **Due Today** *(Amber)*, **In X days** *(Blue)*, **Overdue by X days** *(Red)*, and **Completed** *(Green checkmark)*.

FR-OPS-3 (Labor Legal Compliance & Age Verification):

All workers must be registered in the Employee Registry with full name, national ID, and date of birth. System strictly enforces legal working age:

$$Age = \text{Current Date} - \text{Date of Birth} \geq 18 \text{ Years}$$

FR-OPS-4 (Activity Labor Assignment & Wage Calculation):

Supervisors can assign multiple verified employees to any task, logging hours worked and wage payable amounts:

$$\text{Total Task Wage} = \sum (\text{Hours Worked} \times \text{Hourly Rate})$$

3.5 Harvest Produce & Stock-Controlled Sales (FR-SALE)

FR-SALE-1 (Harvest Produce Intake):

Supervisors log harvested yields specifying crop/produce item name, quantity harvested, and unit of measurement (kg, bags, crates, litres, tonnes).

FR-SALE-2 (Harvest Stock vs. Sales Validation Engine):

The system strictly enforces the business rule that farm produce cannot be sold if insufficient harvest has been recorded for that project:

$$\text{Available Stock} = \sum \text{Harvest Quantity} - \sum \text{Sold Quantity} \geq \text{Requested Sale Quantity}$$

If requested quantity exceeds available harvest stock, the sale transaction is rejected with an explanatory HTTP 400 error.

FR-SALE-3 (Customer Credit & Down-Payment Allocation):

- Sales can be paid in full or on credit terms.
- When $\text{Amount Paid} < \text{Total Amount}$, an existing registered customer **must** be attached to record the debt.

- Partial down-payments are automatically logged as the opening transaction in the customer payment audit trail.
- If a customer's outstanding debt exceeds their `creditLimit`, their status becomes `BLOCKED` and further credit purchases are prohibited.

3.6 Customer Accounts Receivable (AR) & Installment Audit (FR-AR)

FR-AR-1 (Customer Accounts Receivable Ledger):

Each registered customer maintains an ongoing financial ledger:

$$\\$\\text{Outstanding Debt (AR)} = \\text{Total Purchases (Billed)} - \\text{Total Payments (Remitted)}\\$\\$$$

FR-AR-2 (Dedicated Installment Payment Audit Modal):

On `CustomerDetailsPage.jsx`, every sale record contains an **Audit** (`Audit`) button opening the :

- Displays Invoice Total, Amount Paid, Balance Due, and Settlement Badge.
- Chronological table of all installment payments detailing Date, Payment Mode (`MPESA`, `CASH`, `BANK_TRANSFER`, `CHEQUE`), Reference Code, Amount Paid, Balance After, and Notes.
- Quick-action **"Pay Towards This Sale"** button.

FR-AR-3 (Targeted & FIFO Debt Allocation):

- Operators can target installment payments directly to a specific sale invoice (`saleId`).
- If paying general account debt, the system applies payment across the customer's oldest unpaid sales in FIFO order, updating each sale's balance and payment status.

3.7 Supplier Accounts Payable (AP) & Inventory Restock (FR-AP)

FR-AP-1 (Supplier Accounts Payable Ledger):

Each supplier maintains an ongoing accounts payable ledger:

$$\\$\\text{Balance Owed (AP)} = \\text{Total Purchases (Billed)} - \\text{Total Payments (Remitted)}\\$\\$$$

FR-AP-2 (Purchase Invoices & Optional Auto-Restock):

When logging a purchase invoice from a supplier, operators can optionally link a warehouse inventory item. The system automatically increments the warehouse stock level by `restockQuantity`.

FR-AP-3 (Dedicated Purchase Settlement Audit Modal):

On `SupplierDetailsPage.jsx`, every purchase invoice features an **Audit** button opening the :

- Displays Total Billed, Total Paid, Balance Due (AP), and Payment Status.
- Chronological breakdown of all remitted vouchers with payment modes and reference codes.
- Quick-action **"Settle This Invoice"** button.

FR-AP-4 (Targeted & FIFO Voucher Settlement):

Supports settling specific purchase invoices (`purchaseId`) or settling general account debt using FIFO allocation across the oldest unpaid supplier purchases.

3.8 Farm Warehouse Inventory Supplies (FR-INV)

FR-INV-1 (Catalogue & Stock Tracking):

Tracks inventory items across categories (*Seeds*, *Fertilizers*, *Feeds*, *Chemicals*, *Veterinary*, *Tools*) with unit measurement, unit purchase cost, and minimum reorder threshold (`minStockLevel`).

FR-INV-2 (Project Supply Consumption):

Supervisors and managers can log direct supply usage against a project (`POST /inventory/{id}/use`), automatically deducting the quantity from warehouse stock and logging usage notes.

FR-INV-3 (Low Stock Alerting):

Items where $\text{Quantity In Stock} \leq \text{Min Stock Level}$ trigger warning badges on the dashboard and high-priority alerts in the Notification Center.

3.9 Executive Financial Dashboard & Analytics (FR-DASH)**FR-DASH-1 (Ground-Truth Cash & Debt Metrics):**

The Command Center Dashboard computes ground-truth financial KPIs:

- **Cash Received (In Account):** Actual liquid cash collected into farm accounts ($\text{Total Booked Sales} - \text{Uncollected Customer Debt}$).
- **Customer Debts (Uncollected AR):** Total outstanding debt owed by buyers across the enterprise. Features a conditional "**View $\rightarrow$** " button that only appears when $\text{Debt} > 0$.
- **Total Booked Sales:** Gross volume of all recorded crop and livestock sales.
- **Active Projects:** Total ongoing projects with completed project count subtitle.

FR-DASH-2 (Interactive Donut Charts):

- **Revenue by Sector:** Displays revenue share per agricultural sector with percentage hover tooltips.
- **Project Status:** Two-status distribution (**Active** in emerald green vs **Completed** in sky blue) with a background SVG base track ring for elegant zero-state presentation.

FR-DASH-3 (Server-Side Paginated Transactions Feed):

Consolidates sales inflows and supply outflows into a server-side paginated ledger with search, type filters (**ALL** , **SALES** , **SUPPLIES**), and fixed 5-record pagination (**TX_PAGE_SIZE = 5**) sorted by date descending.

FR-DASH-4 (Supervisor Workspace Scoping):

When accessed by a Field Supervisor, the dashboard switches to the scoped **Supervisor Workspace**, showing only their assigned projects, task countdowns, sector tags, and quick inventory shortcuts.

3.10 Real-Time Notification Center (FR-NOTIF)**FR-NOTIF-1 (Proactive Aggregation Engine):**

Continuously monitors:

1. **USERS** : Pending user approval requests.
2. **INVENTORY** : Items at or below minimum stock threshold.
3. **TASKS** : Overdue field tasks and upcoming activities due within 7 days.
4. **FINANCE** : Overdue supplier purchase invoices (AP) and customer debts (AR).

FR-NOTIF-2 (TopBar Interactive Dropdown):

Renders an animated bell icon with live badge count, category filter tabs (**ALL** , **Tasks** , **Stock** , **Users** , **Finance**), and deep links to resolve each notification. Polling runs every 45 seconds and on route transitions.

4. Non-Functional Requirements (NFR)**4.1 Performance & Scalability (NFR-PERF)**

NFR-PERF-1: API endpoint response time under 100ms for 95th percentile under standard enterprise load.

NFR-PERF-2: Database connection pool optimized with **spring.jpa.open-in-view=false** and indexed foreign keys.

NFR-PERF-3: Vite frontend bundle size optimized with tree-shaking, CSS modules, and sub-3-second production builds.

4.2 Security & Data Integrity (NFR-SEC)

NFR-SEC-1: Passwords hashed with BCrypt (strength 10).

NFR-SEC-2: Stateless session validation using `X-User-Id` and `X-User-Role` headers with Spring Security filters.

NFR-SEC-3: Supervisor financial masking at both API and UI presentation layers.

NFR-SEC-4: Prevention of infinite Jackson JSON serialization recursion using `@JsonIgnore` on all bidirectional entity relationships.

4.3 Reliability & Fault Tolerance (NFR-REL)

NFR-REL-1: Email delivery resilience via Resend HTTPS API with SMTP fallback.

NFR-REL-2: Automatic database DDL update with zero destructive schema mutations.

NFR-REL-3: Comprehensive React Error Boundaries with graceful recovery buttons.